Chiral Polyhedra from $\mathrm{AGL}(1, q)$

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DEFINITION

Abstract polyhedra

An *abstract polyhedron* is a ranked poset $\mathcal{P} = (\mathcal{P}, \preccurlyeq)$ satisfying:

(i) There's a least and greatest face.

(ii) Every flag of $\mathcal{P}$ has exactly five faces.

(iii) Each section of $\mathcal{P}$ is flag-connected.

For $F \in \mathcal{P}$ arbitrary with $\mathrm{rk}(F) = i$, write F_i.

(iv) For $0 \leqslant i \leqslant 2$, if $F_{i-1} \prec F_{i+1}$, there are exactly two F_i with

$$F_{i-1} \prec F_i \prec F_{i+1}.$$

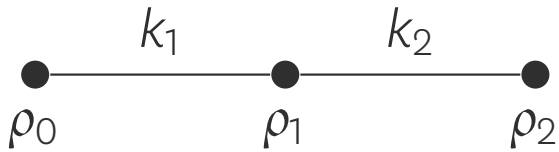
By (iv), for a flag $\Phi \subset \mathcal{P}$ and every $i \in \{0, 1, 2\}$, there is a unique *i-adjacent* flag Φ^i differing only in the i-face.

CHARACTERIZATION

Highly symmetric polyhedra

$\mathcal{P}$ *regular* if $\Gamma(\mathcal{P}) \overset{\text{def}}{=} \mathrm{Aut}(\mathcal{P})$ acts transitively on the flags.

There's a presentation of $\Gamma(\mathcal{P})$ as a *string diagram*:



Highly symmetric polyhedra

If $\Gamma(\mathcal{P})$ induces exactly two orbits on the flags with adjacent flags lying in opposite orbits, then $\mathcal{P}$ *chiral*.

There's a presentation

$$\Gamma(\mathcal{P}) = \left\langle \sigma_1, \sigma_2 \;\middle|\; \sigma_1^{k_1},\ \sigma_2^{k_2},\ (\sigma_1\sigma_2)^2 \right\rangle, \quad \langle\sigma_1\rangle \cap \langle\sigma_2\rangle = 1,$$

but $\Gamma(\mathcal{P})$ admits no *enantiomorphism* $\eta \in \mathrm{Aut}(\Gamma(\mathcal{P}))$:

$$\Gamma(\mathcal{P}_{(\Phi)}) = \langle\sigma_1, \sigma_2\rangle \xleftarrow{\quad\eta\quad} \left\langle \sigma_1^{-1},\ \sigma_1^2\sigma_2 \right\rangle = \Gamma(\mathcal{P}_{(\Phi^0)})\ .$$

In the regular case, $\sigma_i \overset{\text{def}}{=} \rho_{i-1}\rho_i$.

$\Gamma(\mathcal{P}) = [k_1, k_2]^+/N$ for some $N \lhd [k_1, k_2]^+$ not normal in $[k_1, k_2]$.

General affine group

Recall: for a prime power q, the *affine group* is the group of affine maps $\mathbb{F}_q \longrightarrow \mathbb{F}_q$,

$$\mathrm{AGL}(1,q) \stackrel{\text{def}}{=} \left\{ a(a,b) \colon x \longmapsto ax+b \;\middle|\; a \in \mathbb{F}_q^*,\ b \in \mathbb{F}_q \right\}$$

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Fact: when q odd, involutions are $a(-1,c)$.

Chiral polyhedra from $\mathrm{AGL}(1, q)$

Construction:

(i) Pick $\sigma_2 \stackrel{\text{def}}{=} \alpha(a, b)$.

(ii) Pick involution $\tau \stackrel{\text{def}}{=} \sigma_1\sigma_2 = \alpha(-1, c)$.

(iii) Forces $\sigma_1 = \alpha(-\frac{1}{a}, \frac{b}{a} + c)$.

(iv) Set $\Gamma \stackrel{\text{def}}{=} \langle \sigma_1, \sigma_2 \rangle$.

(v) Ensure $\langle \sigma_1 \rangle \cap \langle \sigma_2 \rangle = 1$, *i. e.*, $c \neq \frac{2b}{1-a}$.

Makes regular *or* chiral polyhedron $\mathcal{P}(a, b, c)$ with $\Gamma^+(\mathcal{P}) = \Gamma$.

Chiral polyhedra from $\mathrm{AGL}(1, q)$

Theorem

Fix $a \in \mathbb{F}_q^*$ ($a \neq \pm 1$), $b, c \in \mathbb{F}_q$ ($c \neq \frac{2b}{1-a}$). Write $\mathbb{F}_q^* = \langle a^m \rangle$ for some $1 \leqslant m < q-1$. Let Γ and $\mathcal{P}(a,b,c)$ be as above.

- $\Gamma \leq \mathrm{AGL}(1,q)$ *is the group of a chiral* $\mathcal{P}(a,b,c)$*, or the rotation subgroup of a regular* $\mathcal{P}(a,b,c)$.
- $\mathcal{P}(a,b,c)$ *is of type* $\{o(-\frac{1}{a}), o(a)\}$*, the orders in* $\mathbb{F}_q^*$.
- $\Gamma = \mathrm{AGL}(1,q)$ *iff either* $(q-1,m) = 1$*, or* $(q-1,m) = 2$ *and* $q \equiv 3 \pmod 4$.
- $\mathcal{P}(a,b,c)^\delta = \mathcal{P}(-a, b+ac, c)$.

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Chiral polyhedra from $\mathrm{AGL}(1, q)$

Theorem

Let q be an odd prime power.

- *There are no regular polytopes $\mathcal{P}$ of rank $\geqslant 3$ with $\Gamma(\mathcal{P}) \leq \mathrm{AGL}(1, q)$.*
- *There are no chiral polytopes $\mathcal{P}$ of rank $\geqslant 4$ with $\Gamma(\mathcal{P}) \leq \mathrm{AGL}(1, q)$.*
- *If $\mathcal{P}$ is chiral of type $\{s, t\}$ with $\Gamma(\mathcal{P}) \leq \mathrm{AGL}(1, q)$, then $s, t \mid (q - 1)$.*

APPLICATION

Minimal chiral polyhedra of order $2^n p$

Write $\mathcal{P} \searrow \mathcal{Q}$ (and say $\mathcal{P}$ *covers* $\mathcal{Q}$) iff there is a surjection $\mathcal{P} \to \mathcal{Q}$ preserving ranks of faces and adjacency of flags.

Definition (Minimal polyhedron)

Fix a non-empty collection $\mathscr{C}$ of chiral polyhedra. Say $\mathcal{P} \in \mathscr{C}$ is *minimal* if $\mathcal{P} \searrow \mathcal{Q}$ for some $\mathcal{Q} \in \mathscr{C}$ only when $\mathcal{Q} = \mathcal{P}$.

Obviously, any given $\mathcal{K} \in \mathscr{C}$ covers *at least one* minimal $\mathcal{P} \in \mathscr{C}$.

APPLICATION

Minimal chiral polyhedra of order $2^n p$

Fix $\mathscr{C} = \{\mathcal{P} \mid |\Gamma(\mathcal{P})| = 2^n p$ for some $n > 0$ and odd prime $p\}$.

Theorem

Let $\mathcal{P} \in \mathscr{C}$ be minimal of order $2^n p$. Then $\mathcal{P}$ is of type $\{2^n, 2^n\}$ iff $p \equiv 1 \ (2^n)$. In that case, $n \geqslant 2$, there are 2^{n-2} such $\mathcal{P}$, and

$$\Gamma(\mathcal{P}) \cong \mathbb{F}_p \rtimes \mathbb{Z}/2^n\mathbb{Z} \leq \mathrm{AGL}(1, p).$$

Proposition

Every polyhedron in $\mathscr{C}$ of type $\{2^s, 2^t\}$ covers exactly one minimal polyhedron in $\mathscr{C}$, namely one of those in the above theorem.

APPLICATION

Minimal chiral polyhedra of order $2^n p$

Definition (Mixing operation)

The *mix* of chiral polyhedra $\mathcal{P}, \mathcal{Q}$ is the smallest structure $\mathcal{P} \Diamond \mathcal{Q}$ covering both $\mathcal{P}, \mathcal{Q}$. (So if $\mathcal{K} \searrow \mathcal{P}, \mathcal{Q}$, then $\mathcal{K} \searrow \mathcal{P} \Diamond \mathcal{Q}$.)

Lemma

For $i \in \{1, 2\}$, let $\mathcal{P}_i \in \mathscr{C}$ be minimal of order $2^{n_i} p_i$ where $p_i \equiv 1 \ (2^{n_i})$. If $\mathcal{P}_1 \neq \mathcal{P}_2$, then $p_1 p_2$ divides $|\Gamma(\mathcal{P}_1 \Diamond \mathcal{P}_2)|$.

Proof that $\mathcal{K} \in \mathscr{C}$ of type $\{2^s, 2^t\}$ covers a unique minimal $\mathcal{P} \in \mathscr{C}$.

Suppose $\mathcal{K} \searrow \mathcal{P}_1, \mathcal{P}_2$. Then $\mathcal{K} \searrow \mathcal{P}_1 \Diamond \mathcal{P}_2$. If $\mathcal{P}_1 \neq \mathcal{P}_2$, the lemma gives p^2 divides $|\Gamma(\mathcal{P}_1 \Diamond \mathcal{P}_2)|$, hence p^2 divides $|\Gamma(\mathcal{K})| = 2^n p$. ■

Upshot: All chiral $\mathcal{K}$ of order $2^n p$ and type $\{2^s, 2^t\}$ (the majority*) are uniquely built up from $\mathrm{AGL}(1, p)$ by 2-groups!

FUTURE WORK

Affine polytopes

Generalizations

What enables this particular semidirect product to easily produce the automorphism groups of (chiral) polytopes?

In progress!

Regularity

Regularity seems less common among "affine polytopes." *Why?*

Minimality

Chiral polyhedra from $\mathrm{AGL}(1, q)$ are often minimal wrt $\searrow$, and our generalization (above) is often minimal wrt the # of flags. *Why?*

THANKS!

Questions?

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